

Abhiyash Pratap Singh

East Lansing, MI | 517.204.9389 | abhiyash3101@gmail.com | abhiyash.dev | [GitHub](https://github.com) | [LinkedIn](https://www.linkedin.com)

EDUCATION

Michigan State University - East Lansing, MI
B.S. in Computer Science with Business Cognate

Aug 2021 - Aug 2026

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, FastAPI, Flask, Redux, Tailwind CSS, wxWidgets, Chart.js, NumPy, Pandas, scikit-learn

Infrastructure & Tools: PostgreSQL, SQLite, VictoriaMetrics, Prometheus, Docker, Caddy, Linux, Git, GitLab CI, JWT/HMAC, REST, WebSockets, Agile/Scrum

PROJECTS

Predictive Maintenance Platform | Next.js, TypeScript, Python, PostgreSQL, Docker, scikit-learn Jan 2026 - Aug 2026
Senior Capstone delivered to AkzoNobel Coatings (Warsaw, Indiana)

- Shipped a production predictive-maintenance system to AkzoNobel's Warsaw, IN coatings plant where downtime costs **\$40K-\$60K per hour, per machine**, flagging failures hours before threshold-based alarms would fire.
- Trained survival-analysis and KNN models on vibration/temperature telemetry for daily failure-risk scoring; built the Next.js operator dashboard with role-based auth, Chart.js trends, and PDF/Excel exports.
- Architected a multi-service stack ingesting telemetry into **VictoriaMetrics + PostgreSQL**, containerized with Docker Compose behind Caddy (auto-TLS) with VAPID web-push and SMTP alerting.

Ride-share Microservices Platform | Python, Flask, Docker Compose, SQLite, JWT Nov 2025
Database Systems - individual project

- Split a ride-share backend into **four Flask microservices** (users, availability, reservations, payments) - each with its own SQLite DB, container, and port on a shared Docker network.
- Implemented an **atomic cross-service reservation flow** (JWT auth -> identity check -> payment transfer -> availability teardown) that fails closed so partial failures can't leave money in the wrong account.

CineMatch - Movie Recommender System | React, Vite, FastAPI, SQLite, NumPy, Pandas Apr 2026
Group project - MovieLens dataset (610 users, 9,742 movies, 100K+ ratings)

- Built a full-stack recommender (FastAPI + SQLite + React/Vite) implementing **user-based and item-based collaborative filtering** with Pearson/cosine similarity over mean-centered ratings.
- Evaluated honestly with **RMSE, MAE, Precision@K, and coverage** on a time-aware split - surfaced an accuracy-vs-coverage trade-off (item-based 0.891 RMSE at 81.5% vs user-based 0.905 at 90.8%).

Spartan Touchdown - 2D Game Engine | C++17, wxWidgets, GoogleTest, CMake, Doxygen Fall 2025
CSE 335 - Object-Oriented Software Design (team of 4)

- Architected a level-based 2D game in C++/wxWidgets with **polymorphic entities and data-driven level loading** so new content didn't require code changes - delivered in an Agile/Scrum cadence with a 4-person team.

EXPERIENCE

Undergraduate Learning Assistant - MSU Department of Computer Science Fall 2022

- Supported the lead instructor for **CSE 231 (Intro to Programming, Python)** with 200+ students by grading assignments, holding weekly office hours, and running review sessions.
- Diagnosed root causes of student bugs in 1-on-1 sessions and authored short worked examples on recursion, list/dict semantics, and file I/O.

ACTIVITIES

- Member, **MSU Artificial Intelligence Club** - Sept 2023 to present
- Member, **Kappa Sigma Fraternity** - Feb 2022 to present (Alumni Relations Chair, Award-Winning Rush Chair)